INSE 6450: AI in Systems Engineering

Milestone 1

Due: 11:59 PM (EST), Feb 11, 2026, submit on Moodle.

Include your name and student ID!

- Submit your write-up in PDF (we do not accept hand-written submissions). Text in square brackets are hints that can be ignored.
- All code should be in Python and Pytorch (or Tensorflow).

1: Problem Statement (2 pts)

(1.75 pts) In 150–300 words, clearly define the problem your AI system will solve. Your statement must include:

1. **Task definition**
   - What is the input to your system?
   - What is the output (prediction/decision/recommendation)?
   - Is this classification, regression, ranking, retrieval, forecasting, anomaly detection, etc.?
2. **Target user and use case**
   - Who uses the system and how?
   - What decision will your system support or automate?
3. **Success criteria**
   - What does "good performance" mean in this context?
   - Name at least **1–2 evaluation metrics** you expect to use (even preliminary).

(0.25 pts) Implications for Data Selection: answer the following questions:

- Which constraints most strongly shaped your dataset choice and why?
- Which constraints shaped your sampling/splitting plan?
- Which constraints shaped your feature engineering approach?

2: Data Selection (5 pts)

In Phase 1, you will identify and justify the data needed for your AI system design project, analyze key data risks and constraints (e.g., imbalance, sampling, noise, missingness), and propose an initial feature engineering plan. You must also submit reproducible code that performs your data processing and/or feature engineering.

1. (1 pt) Based on your problem, provide the following:
   - A concise summary of datasets used + how they were obtained and licensed.
   - What each dataset contains (units/records, key fields, labels/targets if applicable)
   - Why each dataset is appropriate for your problem
   - What alternatives you considered and why you didn't choose them (1–2 sentences)
   - If you need to collect your own data, state:
     - collection method, required toold/APIs
     - expected number of samples and feasibility within the course time-frame
2. (1 pt) Answer the following questions regarding data scope and assumptions:
   - Any key assumptions about availability, stationarity, or labeling quality
   - Any known dataset limitations (coverage gaps, biases, proxies, outdatedness, etc.)

3. (1 pt) Report the following information about your data:
   - missingness summary (per column or important fields)
   - summary stats for important numeric fields
   - duplicates/outliers checks
   - example rows or schema table
   - simple plots (histogram, bar chart, correlation heatmap—keep it readable)
4. (2 pts) Identify the following issues and your plan to handle them (if your data does not have any of these issues, explain why):
   - missing values
   - inconsistent formats/units
   - label noise or weak labels
   - outliers and erroneous values
   - duplicate values
   - data drift risk (if temporal)
   - class imbalance
   - labelling (if unsupervised)

3: Feature Engineering (3 pts)

Provide a plan for feature engineering for your data, which includes the following:

1. Candidate features (or feature families) you intend to create
2. Feature types (numeric, categorical encoding, text embeddings, time-window aggregates, graph features, etc.)
3. Any domain-specific transformations (e.g., log scaling, normalization, rolling averages)
4. Handling of categorical variables (one-hot, target encoding, embeddings, etc.)
5. Handling of text/image/audio (tokenization, embedding model choice, resizing, etc.—high-level is fine)

4: Code Submission

You must create a github for your project and share it with the TA and the instructor. Include the following:

- README.md that includes how to run the code, dependencies, and where outputs are saved

Your code should:

- run from raw data
- be clear and commented
- avoid hard-coded absolute paths
- include at least one saved output artifact